NZIF Radio RF Schematic Review — Consolidated Report

Project: NZIF Radio RF
Review Date: 2026-04-11
Sheets Reviewed: audio_subsystem, clock_tree, power_switching_reg, power_ldos, adcs, dacs, rx_mixer, tx_mixer, rx_driver_amps, nzif, dsp, microcontroller
Status: HOLD — Do not proceed to layout freeze until Critical items are resolved

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1. Executive Summary — Top 10 Must-Fix Items

The six domain reviews identified **9 Critical**, **20+ High**, and numerous Moderate findings. The items below represent existential risks to board functionality or component survival and must be resolved before any layout work proceeds.

#	Area	Issue	Risk
1	**RF — Power Symbols**	Wrong lib_ids on all custom rails in rx_mixer and rx_driver_amps (power:+5V/+6V used for −3.5V, +3.3V, +6.5V nets)	Netlist cross-connects rails; −3.5V_RX_OPAMPS shorts to +5V global; ADG884 destruction guaranteed
2	**Power LDOs**	R903/R907/R908/R909 values missing "k" suffix (read as 49.9Ω not 49.9kΩ) → LT3045 SET pins receive 49.9Ω → V_OUT ≈ 5mV on all affected rails	All rails powered by these LDOs produce near-zero voltage; complete subsystem failure
3	**Power Switching**	R1101/R1102 (LT8330): feedback resistors produce −1.79V, not −5V. R1103/R1104 (ADP2303): 100k/100k produces 1.6V, not 4V. Six further resistors have empty Value fields	Multiple supply rails at wrong voltages; dependent circuits receive incorrect supply
4	**Clock Tree**	OSC_CTRL net: R701=1kΩ gives 11kΩ load at AD9523-1 pin 8, below 20kΩ minimum — VCXO control voltage corrupted; PLL will not lock	Radio is inoperable without a locked reference clock
5	**Clock Tree**	U706 part/value mismatch: symbol = SN74CBT3253 (5V), value = SN74CBTLV3253 (LV, 3.6V max) annotated "Rx switches are 5V"	Wrong device assembled; either functionality fails or 3.6V-max device is destroyed at 5V
6	**Precision/DACs**	ADAU1962A (U601/U602): "2.5V" local label may drive AVDD pins — AVDD minimum is 3.0V	DAC operates out of spec or fails to power up; no audio output
7	**Power LDOs**	LT3045 output capacitors at 4.7µF (multiple instances) and LT3094 C931=4.7µF — both below 10µF ceramic minimum for LDO stability	LDO oscillation; noisy or unregulated output on precision analog rails
8	**Audio**	Net label "DVDOUT" should be "DVDDOUT" — potential CM/DVDDOUT net merger causes codec common-mode to be shorted to digital supply	Codec ADAU1361 malfunction; possible device damage
9	**Precision/ADCs**	#PWR0406 uses lib_id power:+3.3V for "+3.3V_CONVERTER" net → AVDD joins global +3.3V net instead of isolated converter rail	ADC analog supply contaminated by digital noise; defeats supply isolation intent
10	**Clock Tree**	AD9523-1 PLL1 external loop filter cap (pin 7, LF1_EXT_CAP) not confirmed present — PLL1 cannot lock	Downstream clocks derived from PLL1 are absent or jitter-dominated

Additional urgent items not in the top 10: RF reconstruction filter absent from tx_mixer sheet; ADA4945-1 feedback caps all DNP (oscillation risk); R910 produces 1.65V not 1.8V on +1.8V_CLOCKTREE rail.

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2. Per-Sheet Findings

2.1 audio_subsystem.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	U1202 (SSM2211S)	5V_LOUDSPEAKER / LOUTP	Speaker- wire at y=152.4/177.8 in proximity to 5V_LOUDSPEAKER at y=157.48 — possible shared node creating supply-to-output short	Verify no junction; inspect netlist for Speaker- node connection to 5V_LOUDSPEAKER
CRITICAL	ADAU1361	DVDDOUT / CM	Net label reads "DVDOUT" — missing second D; may merge CM and DVDDOUT nets	Rename label to "DVDDOUT"; run ERC to confirm nets are distinct
HIGH	U1202 (SSM2211S)	Pin 6 / 5V_LOUDSPEAKER	Only 0.1µF C1214 on 5V supply; peak current ≈400mA requires bulk bypass	Add 10–47µF bulk cap at pin 6
HIGH	ADAU1361	SDA / SCL / IOVDD	I2C pull-up resistors (2kΩ to IOVDD) not confirmed on SDA/SCL in this sheet or parent	Confirm pull-ups exist at exactly one point in hierarchy; add if absent
HIGH	C12xx, R12xx (all)	—	All passive footprints are blank	Assign footprints before layout freeze
MODERATE	ADAU1361	LINP (pin 10) / CM	LINP tied to CM — confirm intentional single-ended input on LINN only	Document design intent; if differential not needed, mark with schematic note
MODERATE	ADAU1361	RINP/RINN/RAUX (pins 12/13/14)	No-connect markers present; datasheet requires connection to CM	Connect all three to CM net or confirm datasheet revision permits NC
MODERATE	C1215, C1216	Headphone output	220µF polarized coupling caps — polarity unverified; fc≈22.6Hz into 32Ω load, marginal	Verify polarity; consider 470µF for fc≈11Hz margin
MODERATE	C1212	SSM2211S BYPASS (pin 2)	0.47µF fitted; datasheet recommends 1µF for PSRR	Change to 1µF
MODERATE	—	Audio inputs	No RF EMI filter caps on any audio input — board is in RF radio environment	Add RC or ferrite+C EMI filter on each audio input trace
MODERATE	U1202 / ADAU1361	LOUTP / LOUTN	Shared between headphone output and SSM2211S input — noise coupling path	Review routing; add isolation if simultaneous use intended
MINOR	ADAU1361	JACKDET/MICIN (pin 4)	No-connect on digital input pin — undefined logic level	Add pull resistor to defined level (IOVDD or GND)
MINOR	ADAU1361	ADDR0 / ADDR1	Both tied GND → I2C address 0x38	Confirm firmware initialises to address 0x38
MINOR	R1201	MIC_IN	49.9kΩ shunt shifts DC bias on MIC_IN — verify operating point is within codec input range	Simulate or calculate DC bias with mic capsule source impedance

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2.2 clock_tree.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	R701 / U702 (AD9523-1)	OSC_CTRL / Pin 8	R701=1kΩ → total load 11kΩ, below 20kΩ minimum per AD9523-1 Table 5 — VCXO VC voltage corrupted	Increase R701 to ≥10kΩ (recommended ≥20kΩ)
CRITICAL	U706 (SN74CBT/LV3253)	VCC / 5V	Symbol lib_id = SN74CBT3253 (5V rated), value = SN74CBTLV3253 (LV, 3.6V max); sheet note says "5V"	Resolve to SN74CBT3253 (5V) on Rx side; update symbol, value, and BOM
HIGH	U702 (AD9523-1)	LF1_EXT_CAP (pin 7)	PLL1 external loop filter capacitor not confirmed present — PLL1 cannot lock	Add loop filter cap per AD9523-1 datasheet Table; confirm net connects to capacitor to GND
HIGH	X701 (CTS 535 TCXO)	EOH / Vc	EOH enable pin connection and Vc control voltage not confirmed; undriven EOH disables oscillator	Confirm EOH driven to enable state; confirm Vc connected to OSC_CTRL or pulled appropriately
HIGH	U702 (AD9523-1)	ZD_IN / ZD_IN_b (pins 70/71)	Zero-delay input pins not confirmed driven or terminated	Drive ZD_IN/ZD_IN_b with reference or terminate per datasheet zero-delay bypass mode
HIGH	X701, U704, U707	—	Footprints missing on CTS 535 TCXO (X701), both SN74LVC8T245 (U704, U707)	Assign footprints; confirm package selection
MODERATE	C730, C732, C734	LDO_PLL1 / LDO_VCO / LDO_DIV_M1	Three 0.47µF bypass caps — verify each individually covers its designated LDO output pin	Confirm one cap per LDO; add if shared or absent
MODERATE	U703 (SN74AUC16374)	OE / 1.8V	OE pull-up to 1.8V not confirmed — undefined state during power-up/down	Add explicit pull-up to 1.8V supply on OE pin
MODERATE	TR701 (ADT1-1)	SMA reference input	No 50Ω primary termination on SMA reference input	Add 50Ω termination at SMA; confirm transformer primary sees 50Ω source
MODERATE	U702 (AD9523-1)	OSC_IN_b (pin 10)	Complementary AC-grounding not confirmed for single-ended clock input mode	Connect OSC_IN_b to AC ground (100nF to GND) per datasheet single-ended input configuration
MODERATE	—	Entire sheet	No frequency plan annotation — no way to verify divider settings by inspection	Add schematic note: reference freq, PLL1/PLL2 VCO freq, all output divider settings, final output frequencies
MODERATE	U705, U706	Mux outputs	Outputs may float when OE_b is high	Add weak pull-down or bus-hold on all mux outputs; confirm OE_b sequencing

2.3 power_switching_reg.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	R1101, R1102	LT8330 (U1101) feedback	1MΩ/806kΩ → V_OUT=−1.79V, not −5V. For −5V with R_lower=806kΩ: R_upper must be ≈4.23MΩ	Replace R1101 with 4.23MΩ (nearest E96: 4.22MΩ); verify formula with selected values
CRITICAL	R1103, R1104	ADP2303 (U1102) feedback	100kΩ/100kΩ → V_OUT=1.6V via V_OUT=0.8×(1+R_TOP/R_BOT), not 4V. For 4V: R_TOP/R_BOT=4; use 400kΩ/100kΩ	Replace R1103 with 400kΩ; recalculate all ADP2303 feedback pairs from first principles
CRITICAL	R1108–R1113	Multiple rails	Six resistors have empty Value fields — output voltages for multiple rails undefined	Calculate and fill all six values before any further review
HIGH	U1102, U1103, U1104 (ADP2303)	PGOOD pins	PGOOD pins have no pull-up resistors and no no-connect markers — power-good function unresolved	Add 100kΩ pull-up to appropriate rail on each PGOOD; or place no-connect and document reason
HIGH	R1108, R1109	LT8330 (U1101) EN/UVLO	Values empty → UVLO threshold undefined	Calculate EN resistor divider for correct UVLO; populate values
HIGH	L1102, L1103, L1104	ADP2303	3.3µH inductors fitted; datasheet recommends 4.7µH — higher ripple current, potential saturation	Change to 4.7µH with adequate saturation current rating; verify DCR
MODERATE	D1104 (PMEG6010CEJ)	—	1A rated device — confirm not placed as catch diode in a 3A path	Verify peak inductor current; uprate diode if needed
MODERATE	Title block	—	Rev field reads "0.-" — typographic error	Correct to proper revision string
MODERATE	Title block	—	Callsign inconsistent with other sheets	Standardise callsign/project name across all title blocks

2.4 power_ldos.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	R903, R907, R908, R909	LT3045 SET pins	Value "49.9" missing "k" suffix → parsed as 49.9Ω → V_OUT≈5mV on all affected rails	Change all four values to "49.9k"; recalculate which resistor sets which rail voltage
CRITICAL	Multiple LT3045 instances	LDO output	Multiple output caps at 4.7µF — LT3045 minimum for stability is 10µF ceramic	Replace with 10µF X5R/X7R 0805 on each LT3045 output; verify MLCC voltage derating
HIGH	R910	+1.8V_CLOCKTREE SET	16.5kΩ → V_OUT=1.65V, not 1.8V. For 1.8V: R_SET=18.0kΩ	Replace R910 with 18.0kΩ (E96: 18.0kΩ)
HIGH	U9xx (LT3045)	+3.3V_CONVERTERS IN	4V input, 3.3V output: only 700mV headroom — marginal at full load and temperature	Review input source; target ≥4.5V input to LT3045 feeding +3.3V_CONVERTERS
HIGH	U904 (LT3094)	Output / C931	C931=4.7µF below 10µF minimum for LT3094 stability	Replace C931 with 10µF
HIGH	All LT3045 instances	SET bypass cap	0.47µF SET bypass cap must be present on each LT3045 SET pin — verify all instances	Check each LT3045 in sheet; confirm bypass cap populates at SET pin
HIGH	C913, C917, C922	SET bypass / LT3045	Value "04.7u" (leading zero) — if intended as 0.47µF (470nF) confirm; if 4.7µF, 10× error on SET bypass function	Clarify intended value; correct notation to "470n" or "0.47u"; re-evaluate PSRR impact
HIGH	All rails	—	No power sequencing: PGOOD not connected to LDO enables; all rails rise simultaneously	Define sequencing requirement; connect PGOOD outputs to downstream LDO EN pins or add sequencer
MODERATE	All LT3045 instances	EN/UV pins	EN/UV left no-connected — no remote shutdown or sequencing capability	If sequencing required, connect EN; otherwise document deliberate NC with rationale
MODERATE	U904 (LT3094)	VIOC (pin 7)	VIOC pin disposition unconfirmed — must not be accidentally driven	Verify pin 7 is explicitly no-connected or properly terminated per datasheet
MODERATE	R911, R912	20mΩ ballast resistors	Implemented as discrete 0603 — datasheet recommends PCB trace for thermal tracking	Consider replacing with PCB trace resistance; if discrete must be used, ensure thermal coupling
MODERATE	—	ADP2303 → LT3045	No ferrite bead between ADP2303 8V output and LT3045 IN pins — switcher noise degrades LDO PSRR at 700kHz	Add ferrite bead (e.g., 600Ω @ 100MHz) + 10µF cap between converter output and LDO input

2.5 adcs.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	#PWR0406	+3.3V_CONVERTER / AVDD	lib_id = power:+3.3V → net joins global +3.3V, not isolated converter rail; AVDD contaminated	Create dedicated power symbol for +3.3V_CONVERTER; do not reuse global power symbol
HIGH	U401 (ADA4807-1)	DISABLE (pin 5)	DISABLE pin connection not confirmed; undriven pin may disable amplifier	Tie DISABLE to V+ (supply rail) to keep amplifier permanently enabled, or connect to control GPIO
HIGH	AIN pairs (all)	AIN+ / AIN−	No differential shunt capacitor across AIN pairs — only 10Ω series resistors; no anti-aliasing Cdiff	Add 100–470pF C0G across each AIN+/AIN− pair at ADC input pins
HIGH	U401, 37+ passives	—	Footprints missing on U401 (ADA4807-1) and 37+ other components	Complete footprint assignment for all components before layout
MODERATE	ADAU1979	IOVDD (pin 12) bypass cap	Decoupling cap returns to AGND — should return to DGND	Reroute bypass cap return to DGND net
MODERATE	ADAU1979	SA_MODE / ADDR0 / ADDR1	Pins not confirmed driven — I2C address undefined; possible floating logic inputs	Tie SA_MODE/ADDR0/ADDR1 to defined levels; document resulting I2C address
MODERATE	R1214	—	Non-sequential refdes — likely copy-paste from another sheet	Renumber R1214 to next available R4xx refdes in this sheet
MINOR	ADAU1979	SDATAOUT2 (pin 14)	Not exported — TDM channel count unclear	Confirm whether TDM4 or TDM8 mode; export SDATAOUT2 if TDM8 is required
MINOR	ADAU1979 / ADAU1962A	MCLKIN / MCLK	Verify MCLKIN for ADC and MCLK for DAC are same domain and frequency	Cross-reference against clock tree output frequencies

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2.6 dacs.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	U601, U602 (ADAU1962A)	AVDD pins / "2.5V"	"2.5V" local label may connect to AVDD — ADAU1962A AVDD minimum is 3.0V	Trace "2.5V" net to all pins; confirm AVDD receives ≥3.0V supply; move 2.5V reference to correct pin
CRITICAL	C623, C624, C628	Decoupling	Value = "0.iu" — typo for "0.1u" (100nF)	Correct all three to "100n" or "0.1u"
HIGH	— (majority of passives)	—	Majority of passive footprints missing	Assign all footprints before layout
MODERATE	R614	IBIAS (3.32kΩ)	No tolerance annotation — 5% substitution would shift IBIAS by up to ±5%	Add "1%" tolerance annotation to R614
MODERATE	Q601 (FZT951TA)	AVDD pass element	VBE drifts −2mV/°C — AVDD supply voltage will shift with temperature	Consider replacing Q601 with an LDO or adding feedback for stable AVDD
MODERATE	C629, C631, C635	10nF filter caps	Dielectric not specified — 10nF X7R will distort in signal path	Specify C0G/NP0 dielectric on all three
MINOR	ADAU1962A (U601, U602)	TDM slot assignments	DSP firmware must map DAC_P0+–DAC_P315± to correct TDM slots	Annotate TDM slot assignments on schematic; cross-reference to DSP firmware

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2.7 rx_mixer.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	All custom power symbols	−3.5V_RX_OPAMPS / +3.3V_RX_MIXER / +6.5V	Wrong lib_ids: power:+5V used for non-5V rails → −3.5V_RX_OPAMPS nets to global +5V in netlist	Replace every power symbol with correct dedicated symbol; regenerate netlist and verify
CRITICAL	SN74CBT3125C	+3.3V_RX_MIXER	5V device on what is labelled as 3.3V rail — out of spec; sheet note says "Rx needs 5V"	If Rx mixer needs 5V, rename rail to +5V_RX_MIXER and supply 5V; update all connected nets
HIGH	HC_201–HC_208	Hold caps / AIN+/AIN−	100nF hold caps give BW≈12kHz with N=4 annotation; should be N=8 giving BW≈16kHz — annotation wrong and cap values may need adjustment	Correct N=8 in bandwidth annotation; recalculate hold cap value for target receive bandwidth; update cap values
MODERATE	ADA4625-2	EP (pin 9) / V−	Exposed pad not confirmed connected to V−	Connect EP to V− (negative supply) as required by datasheet
MODERATE	10µF caps	+6.5V rail	0603 10µF — voltage rating is 6.3V; +6.5V exceeds rating	Replace with 16V-rated caps in 0805 or larger package
MINOR	"HC_" prefix	Capacitor refdes	Non-standard prefix — BOM tools may not parse correctly	Renumber to standard C_xxx refdes or confirm BOM tool handles custom prefixes

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2.8 tx_mixer.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
HIGH	—	Post-mixer output	No reconstruction filter present despite sheet title "16 Phase Tx Harmonic Mixer and Reconstruction Filter"	Locate reconstruction filter in hierarchy or add filter components to this sheet before layout freeze
MODERATE	—	CLK_0/90/135	Duplicate hierarchical label pins at identical coordinates — ERC may miss shorted labels	Clean up duplicate label placements; run ERC and confirm clean
MODERATE	—	phase_N (Rx) vs CLK_N (Tx)	Naming inconsistency for same clock tree outputs — confirm both refer to same physical signals	Standardise net naming across Rx and Tx sheets; annotate which clock tree output drives each
MODERATE	—	Tx DAC differential polarity	Differential polarity per arm not confirmed — incorrect polarity injects image/sideband	Verify LOUTP/LOUTN assignment per Tx arm; annotate polarity on schematic
MINOR	SN74CBTLV3125	Symbol vs value	Value = CBTLV3125 on CBT3125C symbol — assembly will receive wrong device	Create dedicated symbol matching exact part; update BOM
MINOR	U801–U804	Digikey part numbers	Empty — BOM incomplete	Add Digikey/Mouser part numbers for all four

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2.9 rx_driver_amps.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
CRITICAL	All custom power symbols	−3.5V_RX_OPAMPS / +6V	All custom rails use power:+6V as base symbol → −3.5V_RX_OPAMPS shorts to global +6V	Fix all power symbol lib_ids; this is the same root cause as RX-1 — fix together
CRITICAL	ADG884	VDD / "+6V"	ADG884 VDD connected to "+6V" due to DA-1 power symbol error — ADG884 abs max VDD=3.6V	After fixing power symbols, confirm ADG884 VDD connects to correct ≤3.3V rail
HIGH	C343, C322, C319, C320, C340, C341, C344	ADA4945-1 feedback	All 180pF feedback caps DNP in Rev 0.0 — ADA4945-1 with 9.53kΩ Rf will likely oscillate	Either populate feedback caps (recommended) or redesign Rf to stable value without Cf; do not DNP on first prototype
HIGH	BB_0/180/90/270/45/225/135/315	Px_IN+/−	Cross-sheet I/Q differential pairing not confirmed at top level — incorrect pairing destroys I/Q image rejection	Trace every BB_ net to its Px_IN± pin at top level; annotate mapping on nzif.kicad_sch
MODERATE	ADA4945-1 (all 4 instances)	Total supply / DISABLE	10V total supply = rated limit; DISABLE pin disposition not confirmed	Verify supply rails sum to ≤10V under all conditions; confirm DISABLE tied to enable state
MODERATE	LT5400	Resistor pairing / CMRR	Pin pair connectivity must be verified — misconnected pairs ruin CMRR	Verify LT5400 pin-out against KiCad symbol; confirm A1/B1 and A2/B2 pairing correct
MODERATE	ADA4945-1 feedback	9.53kΩ Rf	Thermal noise ≈7.1nV/√Hz dominates amplifier noise floor	Evaluate reducing Rf to 2–4kΩ range; recalculate gain to compensate

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2.10 nzif.kicad_sch / dsp.kicad_sch / microcontroller.kicad_sch

Priority	Ref	Net / Pin	Issue	Action
HIGH	—	TX reconstruction filter	Absent from tx_mixer sheet — must be located or added to hierarchy	Confirm filter location in hierarchy; if missing, design and add before layout freeze
HIGH	HC_201–HC_208	Rx hold caps	100nF + N=8 + R=50Ω → BW≈16kHz — reconcile against target receive bandwidth	Determine target BW; adjust hold cap values; correct N annotation to 8
MODERATE	SWIO_0–3	—	Four SWIO signals leave Microcontroller sheet with no destination or description	Define SWIO destinations; add hierarchical labels or net ties; document purpose
MODERATE	Pico module	XIN	XIN hierarchical label — Pico uses internal crystal; label may create dangling net	Add explicit no-connect to XIN if unused; remove hierarchical label
MODERATE	Q501	USB_RESET / SDP	Schematic note acknowledges Q501 USB_RESET circuit is incomplete	Complete Q501 circuit or remove and document alternative USB reset method before fabrication
MODERATE	CLK_0/90/135 labels	TX mixer	Duplicate hierarchical label pins at identical coordinates	Resolve duplicates; run ERC and confirm zero errors
MODERATE	MCP23017	GPA7 / GPB7	Errata: GPA7/GPB7 must be outputs only — confirm schematic and firmware enforce this	Verify no-connect or output-only assignment on GPA7/GPB7 in schematic; add firmware note
MINOR	SWIO_VLOGIC	—	Level-shifting reference for SWIO bus undocumented	Add net note defining SWIO_VLOGIC voltage and source
MINOR	ADAU1467	DSP_CLK	Clock frequency not annotated — 12.288 or 24.576 MHz?	Add frequency annotation on DSP_CLK net; cross-reference to clock tree output
MINOR	ADAU1979 / ADAU1962A	TDM mode	TDM4 vs TDM8 mode for ADAU1979 not confirmed; two ADAU1962A TDM slot assignments not documented	Annotate TDM mode and slot assignments on schematic; verify against DSP firmware

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3. Prioritized Action List Before Layout Freeze

Actions are grouped by work stream. Items within each tier should be completed in the order listed, as later items may depend on earlier ones.

Tier 1 — Block Layout: Must Fix First (week 1)

#	Action	Owner	Sheets
1	**Fix all RF power symbol lib_ids** in rx_mixer and rx_driver_amps. Replace every power:+5V/+6V used for custom rails with correctly named dedicated symbols. Regenerate netlist and run ERC to confirm zero net cross-connections.	RF	rx_mixer, rx_driver_amps
2	**Correct all LT3045/LT3094 SET resistors**: Add "k" suffix to R903, R907, R908, R909 (49.9Ω → 49.9kΩ); recalculate which resistor value sets each rail; verify LT3094 R903 separately.	Power	power_ldos
3	**Recalculate all switching regulator feedback resistors**: R1101→4.23MΩ (LT8330 for −5V); R1103→400kΩ (ADP2303 for 4V); populate all six empty Value fields (R1108–R1113).	Power	power_switching_reg
4	**Fix OSC_CTRL load resistor**: Increase R701 from 1kΩ to ≥10kΩ to meet AD9523-1 20kΩ minimum load on pin 8.	Clock	clock_tree
5	**Resolve U706 part/value conflict**: Commit to SN74CBT3253 (5V) for Rx switches; update symbol lib_id, value, and BOM entry consistently.	Clock	clock_tree
6	**Fix ADAU1962A AVDD supply**: Trace "2.5V" label to all ADAU1962A pins; redirect AVDD to a ≥3.0V supply rail.	Precision	dacs
7	**Fix #PWR0406 in adcs.kicad_sch**: Replace power:+3.3V symbol with a dedicated +3.3V_CONVERTER power symbol to preserve rail isolation.	Precision	adcs
8	**Rename "DVDOUT" net label to "DVDDOUT"**: Run ERC to confirm CM and DVDDOUT nets are distinct.	Audio	audio_subsystem

Tier 2 — Fix Before PCB Stackup Decisions (week 2)

#	Action	Owner	Sheets
9	**Confirm or add PLL1 loop filter cap** on AD9523-1 pin 7 (LF1_EXT_CAP); add per AD9523-1 datasheet if absent.	Clock	clock_tree
10	**Confirm X701 (CTS 535) EOH and Vc connections**: Drive EOH high to enable; connect Vc to OSC_CTRL net or document alternative.	Clock	clock_tree
11	**Populate LT3045/LT3094 output capacitors**: Replace all 4.7µF instances with ≥10µF ceramic (X5R/X7R, 0805, adequate voltage derating).	Power	power_ldos
12	**Correct R910 to 18.0kΩ** for +1.8V_CLOCKTREE; verify all other LT3045 SET resistor values against target output voltages.	Power	power_ldos
13	**Clarify C913/C917/C922 values** ("04.7u"): confirm intended value (470nF or 4.7µF); correct notation; verify SET pin bypass function.	Power	power_ldos
14	**Populate ADA4945-1 feedback caps** C343/C322/C319/C320/C340/C341/C344 (180pF): do not DNP on first prototype with 9.53kΩ Rf.	RF	rx_driver_amps
15	**Confirm or add Tx reconstruction filter** in hierarchy; if absent from tx_mixer.kicad_sch, design and place filter before layout.	RF/System	tx_mixer, nzif
16	**Correct SN74CBT3125C power rail**: If Rx mixer requires 5V, rename "+3.3V_RX_MIXER" to "+5V_RX_MIXER" and route 5V supply; update all connected power symbols.	RF	rx_mixer
17	**Add ADP2303 PGOOD pull-ups** (100kΩ) on U1102/U1103/U1104; define power sequencing and connect PGOOD to downstream LDO EN pins.	Power	power_switching_reg, power_ldos

Tier 3 — Complete Before Final ERC Sign-Off (week 3)

#	Action	Owner	Sheets
18	**Assign all missing footprints**: U401 (ADA4807-1), X701 (CTS 535), U704/U707 (SN74LVC8T245), all 37+ passives in adcs, majority of passives in dacs, all C12xx/R12xx in audio.	All	all sheets
19	**Add anti-aliasing caps** (100–470pF C0G) across each AIN+/AIN− pair at ADAU1979 input.	Precision	adcs
20	**Confirm I/Q differential pairing** at top level: trace every BB_0/180/90/270/45/225/135/315 net to its Px_IN± destination; annotate mapping in nzif.kicad_sch.	RF/System	nzif, rx_driver_amps
21	**Add U1202 (SSM2211S) bulk bypass cap** (10–47µF) on 5V supply pin 6; add 1µF BYPASS cap C1212.	Audio	audio_subsystem
22	**Correct C623/C624/C628 typo** ("0.iu" → "100n") in dacs.kicad_sch.	Precision	dacs
23	**Verify AD9523-1 ZD_IN/ZD_IN_b** (pins 70/71) driven or terminated per datasheet zero-delay configuration.	Clock	clock_tree
24	**Tie ADA4807-1 DISABLE (pin 5)** to V+ to permanently enable; confirm for all instances.	Precision	adcs
25	**Connect ADA4625-2 EP (pin 9) to V−** in rx_mixer.kicad_sch.	RF	rx_mixer
26	**Replace 10µF 0603 caps on +6.5V rail** with 16V-rated 0805 parts in rx_mixer.kicad_sch.	RF	rx_mixer
27	**Complete Q501 USB_RESET circuit** or document replacement approach; remove incomplete schematic note.	uC	microcontroller
28	**Resolve SWIO_0–3 destinations**: add hierarchical labels, define SWIO_VLOGIC source voltage, and annotate purpose.	uC	microcontroller, nzif
29	**Add I2C pull-up resistors** (2kΩ to IOVDD) on SDA/SCL for ADAU1361 if not present in parent sheet; confirm pull-up at exactly one point in hierarchy.	Audio	audio_subsystem
30	**Add ferrite bead + 10µF cap** between ADP2303 8V output and LT3045 IN pins to restore LDO PSRR at 700kHz.	Power	power_ldos
31	**Add frequency plan annotation** to clock_tree.kicad_sch: reference frequency, PLL1/PLL2 VCO frequencies, all divider settings, output frequencies.	Clock	clock_tree
32	**Run full ERC** with zero errors/warnings as gate criterion for layout freeze; resolve all remaining ERC items not covered above.	All	all sheets

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Sign-off criterion: All Tier 1 actions verified closed, Tier 2 and Tier 3 actions tracked with assigned owners and target dates, before committing to PCB stackup and constraint setup.